# Student ID: S225135766

# Target Grade: HD

# SIT 788- Task 2.1P

### Importing Packages

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

### Importing 2 data sets

```
df_mushroom = pd.read_csv('mushrooms.csv')

df_mushroom.info()
df_mushroom.isnull().sum()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8124 entries, 0 to 8123
Data columns (total 23 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   class                     8124 non-null   object
 1   cap-shape                 8124 non-null   object
 2   cap-surface               8124 non-null   object
 3   cap-color                 8124 non-null   object
 4   bruises                   8124 non-null   object
 5   odor                      8124 non-null   object
 6   gill-attachment           8124 non-null   object
 7   gill-spacing              8124 non-null   object
 8   gill-size                 8124 non-null   object
 9   gill-color                8124 non-null   object
 10  stalk-shape               8124 non-null   object
 11  stalk-root                8124 non-null   object
 12  stalk-surface-above-ring  8124 non-null   object
 13  stalk-surface-below-ring  8124 non-null   object
 14  stalk-color-above-ring    8124 non-null   object
 15  stalk-color-below-ring    8124 non-null   object
 16  veil-type                 8124 non-null   object
 17  veil-color                8124 non-null   object
```

```
 18  ring-number               8124 non-null   object
 19  ring-type                 8124 non-null   object
 20  spore-print-color         8124 non-null   object
 21  population                8124 non-null   object
 22  habitat                   8124 non-null   object
dtypes: object(23)
memory usage: 1.4+ MB

class                       0
cap-shape                   0
cap-surface                 0
cap-color                   0
bruises                     0
odor                        0
gill-attachment             0
gill-spacing                0
gill-size                   0
gill-color                  0
stalk-shape                 0
stalk-root                  0
stalk-surface-above-ring    0
stalk-surface-below-ring    0
stalk-color-above-ring      0
stalk-color-below-ring      0
veil-type                   0
veil-color                  0
ring-number                 0
ring-type                   0
spore-print-color           0
population                  0
habitat                     0
dtype: int64
```

```python
## Importing CKD Dataset
#!pip install ucimlrepo
from ucimlrepo import fetch_ucirepo

# fetch dataset
df_ckd = fetch_ucirepo(id=336)

# metadata
print(df_ckd.metadata)

# variable information
print(df_ckd.variables)
```

```
{'uci_id': 336, 'name': 'Chronic Kidney Disease', 'repository_url':
'https://archive.ics.uci.edu/dataset/336/chronic+kidney+disease', 'data_url':
'https://archive.ics.uci.edu/static/public/336/data.csv', 'abstract': 'This
dataset can be used to predict the chronic kidney disease and it can be
collected from the hospital nearly 2 months of period.', 'area': 'Other',
'tasks': ['Classification'], 'characteristics': ['Multivariate'],
'num_instances': 400, 'num_features': 24, 'feature_types': ['Real'],
'demographics': ['Age'], 'target_col': ['class'], 'index_col': None,
'has_missing_values': 'yes', 'missing_values_symbol': 'NaN',
```

'year_of_dataset_creation': 2015, 'last_updated': 'Mon Mar 04 2024', 'dataset_doi': '10.24432/C5G020', 'creators': ['L. Rubini', 'P. Soundarapandian', 'P. Eswaran'], 'intro_paper': None, 'additional_info': {'summary': 'We use the following representation to collect the dataset\r\n age\t\t-\tage\t\r\n\t\t\tbp\t\t-\tblood pressure\r\n\t\t\tsg\t\t-\tspecific gravity\r\n\t\t\tal\t\t-  \talbumin\r\n\t\t\tsu\t\t-\tsugar\r\n\t\t\trbc\t\t-\tred blood cells\r\n\t\t\tpc\t\t-\tpus cell\r\n\t\t\tpcc\t\t-\tpus cell clumps\r\n\t\t\tba\t\t-\tbacteria\r\n\t\t\tbgr\t\t-\tblood glucose random\r\n\t\t\tbu\t\t-\tblood urea\r\n\t\t\tsc\t\t-\tserum creatinine\r\n\t\t\tsod\t\t-\tsodium\r\n\t\t\tpot\t\t-\tpotassium\r\n\t\t\themo\t\t-\themoglobin\r\n\t\t\tpcv\t\t-\tpacked cell volume\r\n\t\t\twc\t\t-\twhite blood cell count\r\n\t\t\trc\t\t-\tred blood cell count\r\n\t\t\thtn\t\t-\thypertension\r\n\t\t\tdm\t\t-\tdiabetes mellitus\r\n\t\t\tcad\t\t-\tcoronary artery disease\r\n\t\t\tappet\t\t-\tappetite\r\n\t\t\tpe\t\t-\tpedal edema\r\n\t\t\tane\t\t-\tanemia\r\n\t\t\tclass\t\t-\tclass\t', 'purpose': None, 'funded_by': None, 'instances_represent': None, 'recommended_data_splits': None, 'sensitive_data': None, 'preprocessing_description': None, 'variable_info': 'We use 24 + class = 25 ( 11  numeric ,14  nominal)\r\n1.Age(numerical)\r\n  \t  \tage in years\r\n \t2.Blood Pressure(numerical)\r\n\t       \tbp in mm/Hg\r\n \t3.Specific Gravity(nominal)\r\n\t  \tsg - (1.005,1.010,1.015,1.020,1.025)\r\n \t4.Albumin(nominal)\r\n\t\tal - (0,1,2,3,4,5)\r\n \t5.Sugar(nominal)\r\n\t\tsu - (0,1,2,3,4,5)\r\n \t6.Red Blood Cells(nominal)\r\n\t\trbc - (normal,abnormal)\r\n \t7.Pus Cell (nominal)\r\n\t\tpc - (normal,abnormal)\r\n \t8.Pus Cell clumps(nominal)\r\n\t\tpcc - (present,notpresent)\r\n \t9.Bacteria(nominal)\r\n\t\tba  - (present,notpresent)\r\n \t10.Blood Glucose Random(numerical)\t\t\r\n\t\tbgr in mgs/dl\r\n \t11.Blood Urea(numerical)\t\r\n\t\tbu in mgs/dl\r\n \t12.Serum Creatinine(numerical)\t\r\n\t\tsc in mgs/dl\r\n \t13.Sodium(numerical)\r\n\t\tsod in mEq/L\r\n \t14.Potassium(numerical)\t\r\n\t\tpot in mEq/L\r\n \t15.Hemoglobin(numerical)\r\n\t\themo in gms\r\n \t16.Packed  Cell Volume(numerical)\r\n \t17.White Blood Cell Count(numerical)\r\n\t\twc in cells/cumm\r\n \t18.Red Blood Cell Count(numerical)\t\r\n\t\trc in millions/cmm\r\n \t19.Hypertension(nominal)\t\r\n\t\thtn - (yes,no)\r\n \t20.Diabetes Mellitus(nominal)\t\r\n\t\tdm - (yes,no)\r\n \t21.Coronary Artery Disease(nominal)\r\n\t\tcad - (yes,no)\r\n \t22.Appetite(nominal)\t\r\n\t\tappet - (good,poor)\r\n \t23.Pedal Edema(nominal)\r\n\t\tpe - (yes,no)\t\r\n \t24.Anemia(nominal)\r\n\t\tane - (yes,no)\r\n \t25.Class (nominal)\t\t\r\n\t\tclass - (ckd,notckd)', 'citation': None}}

| | name | role | type | demographic | description | \ |
|---|---|---|---|---|---|---|
| 0 | age | Feature | Integer | Age | None | |
| 1 | bp | Feature | Integer | None | blood pressure | |
| 2 | sg | Feature | Categorical | None | specific gravity | |
| 3 | al | Feature | Categorical | None | albumin | |
| 4 | su | Feature | Categorical | None | sugar | |
| 5 | rbc | Feature | Binary | None | red blood cells | |
| 6 | pc | Feature | Binary | None | pus cell | |
| 7 | pcc | Feature | Binary | None | pus cell clumps | |
| 8 | ba | Feature | Binary | None | bacteria | |
| 9 | bgr | Feature | Integer | None | blood glucose random | |
| 10 | bu | Feature | Integer | None | blood urea | |
| 11 | sc | Feature | Continuous | None | serum creatinine | |

| | | | | | |
|---|---|---|---|---|---|
| 12 | sod | Feature | Integer | None | sodium |
| 13 | pot | Feature | Continuous | None | potassium |
| 14 | hemo | Feature | Continuous | None | hemoglobin |
| 15 | pcv | Feature | Integer | None | packed cell volume |
| 16 | wbcc | Feature | Integer | None | white blood cell count |
| 17 | rbcc | Feature | Continuous | None | red blood cell count |
| 18 | htn | Feature | Binary | None | hypertension |
| 19 | dm | Feature | Binary | None | diabetes mellitus |
| 20 | cad | Feature | Binary | None | coronary artery disease |
| 21 | appet | Feature | Binary | None | appetite |
| 22 | pe | Feature | Binary | None | pedal edema |
| 23 | ane | Feature | Binary | None | anemia |
| 24 | class | Target | Binary | None | ckd or not ckd |

| | units | missing_values |
|---|---|---|
| 0 | year | yes |
| 1 | mm/Hg | yes |
| 2 | None | yes |
| 3 | None | yes |
| 4 | None | yes |
| 5 | None | yes |
| 6 | None | yes |
| 7 | None | yes |
| 8 | None | yes |
| 9 | mgs/dl | yes |
| 10 | mgs/dl | yes |
| 11 | mgs/dl | yes |
| 12 | mEq/L | yes |
| 13 | mEq/L | yes |
| 14 | gms | yes |
| 15 | None | yes |
| 16 | cells/cmm | yes |
| 17 | millions/cmm | yes |
| 18 | None | yes |
| 19 | None | yes |
| 20 | None | yes |
| 21 | None | yes |
| 22 | None | yes |
| 23 | None | yes |
| 24 | None | no |

# Data set Description & Problem Statement

## 1. CKD Data Set

The dataset comprises 25 columns, 24 features and one target variable (class). Features include 11 numerical measurements.A significant data quality challenge was identified during pre-analysis: the dataset contains 1,013 missing cells across 24 of 25 columns, representing 10.1% of all data. The rbc column alone has 38% missing values. Additional issues include tab/space-polluted categorical values (e.g., '\tno', ' yes' in the dm column),

one corrupted class label ('no' instead of 'ckd'/'notckd'), and three ARFF rows with a trailing empty column. These issues were systematically identified and resolved during preprocessing.

The classification problem involves predicting whether a patient has chronic kidney disease (ckd) or does not (notckd) based on 24 clinical features. Early and accurate diagnosis of CKD is clinically critical, the disease often progresses silently until it reaches an advanced stage. A model with ***high recall for the positive (CKD) class*** is therefore essential, as ***false negatives (missing a sick patient) carry greater clinical risk than false positives***.

### 2. Mushroom Data Set

The Mushroom Classification dataset was sourced from Kaggle (UCI Machine Learning repository). It contains 8,124 records describing physical characteristics of mushrooms from 23 species of the Agaricus and Lepiota family. Each mushroom is labelled as either edible (e) or poisonous (p). The dataset contains 23 columns, 22 features and one target variable. All features are nominal categorical variables encoded as single-letter codes.

The veil-type column contains zero variance (every row equals 'p') and was removed before modelling, and the stalk-root column uses '?' to encode missing values rather than leaving cells blank.

The classification problem involves predicting mushroom edibility from physical characteristics alone. This is a safety-critical task; consuming a poisonous mushroom can be fatal. As with CKD, ***recall for the positive (poisonous) class is the priority metric***, since ***failing to identify a poisonous mushroom as dangerous (false negative) is far more consequential*** than incorrectly classifying an edible mushroom as poisonous (false positive).

# Data set Comparison

| Characteristic | CKD Dataset | Mushroom Dataset |
|---|---|---|
| **Rows** | 400 (small) | 8,124 (large) |
| **Features** | 24 (mixed types) | 21 (all categorical) |
| **Feature Types** | 11 numeric, 14 categorical | All nominal categorical |
| **Missing Values** | YES, up to 38% in rbc | 1 column (stalk-root uses '?') |
| **Class Balance** | 63% CKD / 37% notCKD | 52% edible / 48% poisonous |
| **Data Quality Issues** | Tabs, corrupted labels | Zero-variance column, '?' missing |

# ML Pipeline

| Stage | Process | Key Actions |
|---|---|---|
| **1. Data Ingestion** | Load CKD ARFF / Mushroom CSV | Convert `?` → `NaN` |
| **2. Data Preparation** | Strip whitespace / Fix labels | Impute missing values & Label encode |
| **3. Train/Test Split** | 80% Train / 20% Test | `stratify=y`, `random_state=42` |
| **4. Model Training** | DecisionTree & RandomForest | Use `gini` criterion for both |
| **5. Model Evaluation** | Performance Metrics | Accuracy, F1-Score, Confusion Matrix, 5-Fold CV |
| **6. Model Export** | `pickle.dump()` | Save best model as `.pkl` |

## EDA - Mushroom Data Set ( df_mushroom)

```python
# Store a copy of the Mushrooms data set

df_m1 = df_mushroom.copy()

print(df_m1.describe())
print(df_m1.isnull().sum())
print(df_m1.nunique())
```

```
       class cap-shape cap-surface cap-color bruises  odor gill-attachment  \
count   8124      8124        8124      8124    8124  8124            8124
unique     2         6           4        10       2     9               2
top        e         x           y         n       f     n               f
freq    4208      3656        3244      2284    4748  3528            7914

       gill-spacing gill-size gill-color  ... stalk-surface-below-ring  \
count          8124      8124       8124  ...                     8124
unique            2         2         12  ...                        4
top               c         b          b  ...                        s
freq           6812      5612       1728  ...                     4936

       stalk-color-above-ring stalk-color-below-ring veil-type veil-color  \
count                    8124                   8124      8124       8124
unique                      9                      9         1          4
top                         w                      w         p          w
freq                     4464                   4384      8124       7924

       ring-number ring-type spore-print-color population habitat
count         8124      8124              8124       8124    8124
unique           3         5                 9          6       7
top              o         p                 w          v       d
freq          7488      3968              2388       4040    3148

[4 rows x 23 columns]
class                     0
cap-shape                 0
cap-surface               0
```

```
cap-color                   0
bruises                     0
odor                        0
gill-attachment             0
gill-spacing                0
gill-size                   0
gill-color                  0
stalk-shape                 0
stalk-root                  0
stalk-surface-above-ring    0
stalk-surface-below-ring    0
stalk-color-above-ring      0
stalk-color-below-ring      0
veil-type                   0
veil-color                  0
ring-number                 0
ring-type                   0
spore-print-color           0
population                  0
habitat                     0
dtype: int64
class                       2
cap-shape                   6
cap-surface                 4
cap-color                  10
bruises                     2
odor                        9
gill-attachment             2
gill-spacing                2
gill-size                   2
gill-color                 12
stalk-shape                 2
stalk-root                  5
stalk-surface-above-ring    4
stalk-surface-below-ring    4
stalk-color-above-ring      9
stalk-color-below-ring      9
veil-type                   1
veil-color                  4
ring-number                 3
ring-type                   5
spore-print-color           9
population                  6
habitat                     7
dtype: int64
```

## Data Preprocessing - Mushrooms dataset

```python
#Check unique values for object data types
for col in df_m1.select_dtypes(include='object').columns:
    n = df_m1[col].nunique()
    vals = sorted(df_m1[col].dropna().unique())
    print(f"{col:10} | unique: {n:3} | values: {vals}")
```

```
class      | unique:   2 | values: ['e', 'p']
cap-shape  | unique:   6 | values: ['b', 'c', 'f', 'k', 's', 'x']
cap-surface | unique:   4 | values: ['f', 'g', 's', 'y']
cap-color  | unique:  10 | values: ['b', 'c', 'e', 'g', 'n', 'p', 'r', 'u',
'w', 'y']
bruises    | unique:   2 | values: ['f', 't']
odor       | unique:   9 | values: ['a', 'c', 'f', 'l', 'm', 'n', 'p', 's',
'y']
gill-attachment | unique:   2 | values: ['a', 'f']
gill-spacing | unique:   2 | values: ['c', 'w']
gill-size  | unique:   2 | values: ['b', 'n']
gill-color | unique:  12 | values: ['b', 'e', 'g', 'h', 'k', 'n', 'o', 'p',
'r', 'u', 'w', 'y']
stalk-shape | unique:   2 | values: ['e', 't']
stalk-root | unique:   5 | values: ['?', 'b', 'c', 'e', 'r']
stalk-surface-above-ring | unique:   4 | values: ['f', 'k', 's', 'y']
stalk-surface-below-ring | unique:   4 | values: ['f', 'k', 's', 'y']
stalk-color-above-ring | unique:   9 | values: ['b', 'c', 'e', 'g', 'n', 'o',
'p', 'w', 'y']
stalk-color-below-ring | unique:   9 | values: ['b', 'c', 'e', 'g', 'n', 'o',
'p', 'w', 'y']
veil-type  | unique:   1 | values: ['p']
veil-color | unique:   4 | values: ['n', 'o', 'w', 'y']
ring-number | unique:   3 | values: ['n', 'o', 't']
ring-type  | unique:   5 | values: ['e', 'f', 'l', 'n', 'p']
spore-print-color | unique:   9 | values: ['b', 'h', 'k', 'n', 'o', 'r', 'u',
'w', 'y']
population | unique:   6 | values: ['a', 'c', 'n', 's', 'v', 'y']
habitat    | unique:   7 | values: ['d', 'g', 'l', 'm', 'p', 'u', 'w']
```

```python
# Since Stoalk-root uses '?' for unknown, we need to convert it to 'Nan'

df_m1['stalk-root'] = df_m1['stalk-root'].replace('?',np.nan)

#Since the veil-type have only one value which is 'p' it will not directly
         contibute to the model. SO we can drop it

df_m1.drop(columns=['veil-type'], inplace = True)

for col in df_m1.select_dtypes(include='object').columns:
    n = df_m1[col].nunique()
    vals = sorted(df_m1[col].dropna().unique())
    print(f"{col:10} | unique: {n:3} | values: {vals}")
```

```
class      | unique:   2 | values: ['e', 'p']
cap-shape  | unique:   6 | values: ['b', 'c', 'f', 'k', 's', 'x']
cap-surface | unique:   4 | values: ['f', 'g', 's', 'y']
cap-color  | unique:  10 | values: ['b', 'c', 'e', 'g', 'n', 'p', 'r', 'u',
'w', 'y']
bruises    | unique:   2 | values: ['f', 't']
odor       | unique:   9 | values: ['a', 'c', 'f', 'l', 'm', 'n', 'p', 's',
'y']
gill-attachment | unique:   2 | values: ['a', 'f']
gill-spacing | unique:   2 | values: ['c', 'w']
gill-size  | unique:   2 | values: ['b', 'n']
gill-color | unique:  12 | values: ['b', 'e', 'g', 'h', 'k', 'n', 'o', 'p',
```

```
'r', 'u', 'w', 'y']
stalk-shape | unique:   2 | values: ['e', 't']
stalk-root | unique:   4 | values: ['b', 'c', 'e', 'r']
stalk-surface-above-ring | unique:   4 | values: ['f', 'k', 's', 'y']
stalk-surface-below-ring | unique:   4 | values: ['f', 'k', 's', 'y']
stalk-color-above-ring | unique:   9 | values: ['b', 'c', 'e', 'g', 'n', 'o',
'p', 'w', 'y']
stalk-color-below-ring | unique:   9 | values: ['b', 'c', 'e', 'g', 'n', 'o',
'p', 'w', 'y']
veil-color | unique:   4 | values: ['n', 'o', 'w', 'y']
ring-number | unique:   3 | values: ['n', 'o', 't']
ring-type  | unique:   5 | values: ['e', 'f', 'l', 'n', 'p']
spore-print-color | unique:   9 | values: ['b', 'h', 'k', 'n', 'o', 'r', 'u',
'w', 'y']
population | unique:   6 | values: ['a', 'c', 'n', 's', 'v', 'y']
habitat    | unique:   7 | values: ['d', 'g', 'l', 'm', 'p', 'u', 'w']
```

```
print(((df_m1.isnull().sum()/len(df_m1))*100))
```

```
class                        0.000000
cap-shape                    0.000000
cap-surface                  0.000000
cap-color                    0.000000
bruises                      0.000000
odor                         0.000000
gill-attachment              0.000000
gill-spacing                 0.000000
gill-size                    0.000000
gill-color                   0.000000
stalk-shape                  0.000000
stalk-root                  30.526834
stalk-surface-above-ring     0.000000
stalk-surface-below-ring     0.000000
stalk-color-above-ring       0.000000
stalk-color-below-ring       0.000000
veil-color                   0.000000
ring-number                  0.000000
ring-type                    0.000000
spore-print-color            0.000000
population                   0.000000
habitat                      0.000000
dtype: float64
```

```
print(df_m1['stalk-root'].unique())

# We will use Mode, to impute the nan value

mode_val = df_m1['stalk-root'].mode()[0]

df_m1['stalk-root']= df_m1['stalk-root'].fillna(mode_val)

#Check remaining missing values
print(((df_m1.isnull().sum()/len(df_m1))*100))
```

```
['e' 'c' 'b' 'r' nan]
class                       0.0
```

```
cap-shape                   0.0
cap-surface                 0.0
cap-color                   0.0
bruises                     0.0
odor                        0.0
gill-attachment             0.0
gill-spacing                0.0
gill-size                   0.0
gill-color                  0.0
stalk-shape                 0.0
stalk-root                  0.0
stalk-surface-above-ring    0.0
stalk-surface-below-ring    0.0
stalk-color-above-ring      0.0
stalk-color-below-ring      0.0
veil-color                  0.0
ring-number                 0.0
ring-type                   0.0
spore-print-color           0.0
population                  0.0
habitat                     0.0
dtype: float64
```

```python
## Encoding target value - "Class" for Classification

## Encode all columns in the mushroom dataset
from sklearn.preprocessing import LabelEncoder

mushroom_encoder={}

for col in df_m1.columns:
    le = LabelEncoder()
    df_m1[col] = le.fit_transform(df_m1[col].astype(str))
    mushroom_encoder[col] = le

df_m1.head()
```

| | class | cap-shape | cap-surface | cap-color | bruises | odor | gill-attachment | gill-spacing | gill-size |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 5 | 2 | 4 | 1 | 6 | 1 | 0 | 1 |
| **1** | 0 | 5 | 2 | 9 | 1 | 0 | 1 | 0 | 0 |
| **2** | 0 | 0 | 2 | 8 | 1 | 3 | 1 | 0 | 0 |
| **3** | 1 | 5 | 3 | 8 | 1 | 6 | 1 | 0 | 1 |
| **4** | 0 | 5 | 2 | 3 | 0 | 5 | 1 | 1 | 0 |

5 rows × 22 columns

# EDA - CKD Data set ( df_ckd)

```
## Store a copy of the ckd data set

df_repo = fetch_ucirepo(id=336)
print(df_repo.metadata)
print(df_repo.variables)

x = df_repo.data.features
y = df_repo.data.targets

df_ckd1 = pd.concat([x, y], axis=1)

print(df_ckd1.shape)
print(df_ckd1.isnull().sum())
print(df_ckd1.head())
```


```
{'uci_id': 336, 'name': 'Chronic Kidney Disease', 'repository_url':
'https://archive.ics.uci.edu/dataset/336/chronic+kidney+disease', 'data_url':
'https://archive.ics.uci.edu/static/public/336/data.csv', 'abstract': 'This
dataset can be used to predict the chronic kidney disease and it can be
collected from the hospital nearly 2 months of period.', 'area': 'Other',
'tasks': ['Classification'], 'characteristics': ['Multivariate'],
'num_instances': 400, 'num_features': 24, 'feature_types': ['Real'],
'demographics': ['Age'], 'target_col': ['class'], 'index_col': None,
'has_missing_values': 'yes', 'missing_values_symbol': 'NaN',
'year_of_dataset_creation': 2015, 'last_updated': 'Mon Mar 04 2024',
'dataset_doi': '10.24432/C5G020', 'creators': ['L. Rubini', 'P.
Soundarapandian', 'P. Eswaran'], 'intro_paper': None, 'additional_info':
{'summary': 'We use the following representation to collect the dataset\r\n
age\t\t-\tage\t\r\n\t\t\tbp\t\t-\tblood pressure\r\n\t\t\tsg\t\t-\tspecific
gravity\r\n\t\t\tal\t\t-   \talbumin\r\n\t\t\tsu\t\t-
\tsugar\r\n\t\t\trbc\t\t-\tred blood cells\r\n\t\t\tpc\t\t-\tpus
cell\r\n\t\t\tpcc\t\t-\tpus cell clumps\r\n\t\t\tba\t\t-
\tbacteria\r\n\t\t\tbgr\t\t-\tblood glucose random\r\n\t\t\tbu\t\t-\tblood
urea\r\n\t\t\tsc\t\t-\tserum creatinine\r\n\t\t\tsod\t\t-
\tsodium\r\n\t\t\tpot\t\t-\tpotassium\r\n\t\t\themo\t\t-
\themoglobin\r\n\t\t\tpcv\t\t-\tpacked cell volume\r\n\t\t\twc\t\t-\twhite
blood cell count\r\n\t\t\trc\t\t-\tred blood cell count\r\n\t\t\thtn\t\t-
\thypertension\r\n\t\t\tdm\t\t-\tdiabetes mellitus\r\n\t\t\tcad\t\t-
\tcoronary artery disease\r\n\t\t\tappet\t\t-\tappetite\r\n\t\t\tpe\t\t-
\tpedal edema\r\n\t\t\tane\t\t-\tanemia\r\n\t\t\tclass\t\t-\tclass\t',
'purpose': None, 'funded_by': None, 'instances_represent': None,
'recommended_data_splits': None, 'sensitive_data': None,
'preprocessing_description': None, 'variable_info': 'We use 24 + class = 25 (
11  numeric ,14  nominal)\r\n1.Age(numerical)\r\n  \t  \tage in years\r\n
\t2.Blood Pressure(numerical)\r\n\t       \tbp in mm/Hg\r\n \t3.Specific
Gravity(nominal)\r\n\t  \tsg - (1.005,1.010,1.015,1.020,1.025)\r\n
\t4.Albumin(nominal)\r\n\t\tal - (0,1,2,3,4,5)\r\n
\t5.Sugar(nominal)\r\n\t\tsu - (0,1,2,3,4,5)\r\n \t6.Red Blood
Cells(nominal)\r\n\t\trbc - (normal,abnormal)\r\n \t7.Pus Cell
(nominal)\r\n\t\tpc - (normal,abnormal)\r\n \t8.Pus Cell
clumps(nominal)\r\n\t\tpcc - (present,notpresent)\r\n
\t9.Bacteria(nominal)\r\n\t\tba  - (present,notpresent)\r\n \t10.Blood
Glucose Random(numerical)\t\t\r\n\t\tbgr in mgs/dl\r\n \t11.Blood
```

Urea(numerical)\t\r\n\t\tbu in mgs/dl\r\n \t12.Serum Creatinine(numerical)\t\r\n\t\tsc in mgs/dl\r\n \t13.Sodium(numerical)\r\n\t\tsod in mEq/L\r\n \t14.Potassium(numerical)\t\r\n\t\tpot in mEq/L\r\n \t15.Hemoglobin(numerical)\r\n\t\themo in gms\r\n \t16.Packed  Cell Volume(numerical)\r\n \t17.White Blood Cell Count(numerical)\r\n\t\twc in cells/cumm\r\n \t18.Red Blood Cell Count(numerical)\t\r\n\t\trc in millions/cmm\r\n \t19.Hypertension(nominal)\t\r\n\t\thtn - (yes,no)\r\n \t20.Diabetes Mellitus(nominal)\t\r\n\t\tdm - (yes,no)\r\n \t21.Coronary Artery Disease(nominal)\r\n\t\tcad - (yes,no)\r\n \t22.Appetite(nominal)\t\r\n\t\tappet - (good,poor)\r\n \t23.Pedal Edema(nominal)\r\n\t\tpe - (yes,no)\t\r\n \t24.Anemia(nominal)\r\n\t\tane - (yes,no)\r\n \t25.Class (nominal)\t\t\r\n\t\tclass - (ckd,notckd)', 'citation': None}}

| | name | role | type | demographic | description | \ |
|---|---|---|---|---|---|---|
| 0 | age | Feature | Integer | Age | None | |
| 1 | bp | Feature | Integer | None | blood pressure | |
| 2 | sg | Feature | Categorical | None | specific gravity | |
| 3 | al | Feature | Categorical | None | albumin | |
| 4 | su | Feature | Categorical | None | sugar | |
| 5 | rbc | Feature | Binary | None | red blood cells | |
| 6 | pc | Feature | Binary | None | pus cell | |
| 7 | pcc | Feature | Binary | None | pus cell clumps | |
| 8 | ba | Feature | Binary | None | bacteria | |
| 9 | bgr | Feature | Integer | None | blood glucose random | |
| 10 | bu | Feature | Integer | None | blood urea | |
| 11 | sc | Feature | Continuous | None | serum creatinine | |
| 12 | sod | Feature | Integer | None | sodium | |
| 13 | pot | Feature | Continuous | None | potassium | |
| 14 | hemo | Feature | Continuous | None | hemoglobin | |
| 15 | pcv | Feature | Integer | None | packed cell volume | |
| 16 | wbcc | Feature | Integer | None | white blood cell count | |
| 17 | rbcc | Feature | Continuous | None | red blood cell count | |
| 18 | htn | Feature | Binary | None | hypertension | |
| 19 | dm | Feature | Binary | None | diabetes mellitus | |
| 20 | cad | Feature | Binary | None | coronary artery disease | |
| 21 | appet | Feature | Binary | None | appetite | |
| 22 | pe | Feature | Binary | None | pedal edema | |
| 23 | ane | Feature | Binary | None | anemia | |
| 24 | class | Target | Binary | None | ckd or not ckd | |

| | units | missing_values |
|---|---|---|
| 0 | year | yes |
| 1 | mm/Hg | yes |
| 2 | None | yes |
| 3 | None | yes |
| 4 | None | yes |
| 5 | None | yes |
| 6 | None | yes |
| 7 | None | yes |
| 8 | None | yes |
| 9 | mgs/dl | yes |
| 10 | mgs/dl | yes |
| 11 | mgs/dl | yes |
| 12 | mEq/L | yes |
| 13 | mEq/L | yes |

```
14           gms            yes
15          None            yes
16     cells/cmm            yes
17  millions/cmm            yes
18          None            yes
19          None            yes
20          None            yes
21          None            yes
22          None            yes
23          None            yes
24          None             no
(400, 25)
age        9
bp        12
sg        47
al        46
su        49
rbc      152
pc        65
pcc        4
ba         4
bgr       44
bu        19
sc        17
sod       87
pot       88
hemo      52
pcv       71
wbcc     106
rbcc     131
htn        2
dm         2
cad        2
appet      1
pe         1
ane        1
class      0
dtype: int64
    age    bp     sg   al   su     rbc        pc         pcc          ba  \
0  48.0  80.0  1.020  1.0  0.0     NaN    normal  notpresent  notpresent
1   7.0  50.0  1.020  4.0  0.0     NaN    normal  notpresent  notpresent
2  62.0  80.0  1.010  2.0  3.0  normal    normal  notpresent  notpresent
3  48.0  70.0  1.005  4.0  0.0  normal  abnormal     present  notpresent
4  51.0  80.0  1.010  2.0  0.0  normal    normal  notpresent  notpresent

     bgr  ...   pcv    wbcc  rbcc  htn   dm  cad  appet   pe  ane class
0  121.0  ...  44.0  7800.0   5.2  yes  yes   no   good   no   no   ckd
1    NaN  ...  38.0  6000.0   NaN   no   no   no   good   no   no   ckd
2  423.0  ...  31.0  7500.0   NaN   no  yes   no   poor   no  yes   ckd
3  117.0  ...  32.0  6700.0   3.9  yes   no   no   poor  yes  yes   ckd
4  106.0  ...  35.0  7300.0   4.6   no   no   no   good   no   no   ckd

[5 rows x 25 columns]
```

```
# Data Precheck
```

```python
for col in df_ckd1.select_dtypes(include='object').columns:
    n = df_ckd1[col].nunique()
    vals = sorted(df_ckd1[col].dropna().unique())
    print(f"{col:10} | unique: {n:3} | values: {vals}")


#After inspecting the object datatypes, we can see 'dm', and 'class' features
         have some erros we need to corrected.
```

```
rbc        | unique:   2 | values: ['abnormal', 'normal']
pc         | unique:   2 | values: ['abnormal', 'normal']
pcc        | unique:   2 | values: ['notpresent', 'present']
ba         | unique:   2 | values: ['notpresent', 'present']
htn        | unique:   2 | values: ['no', 'yes']
dm         | unique:   3 | values: ['\tno', 'no', 'yes']
cad        | unique:   2 | values: ['no', 'yes']
appet      | unique:   2 | values: ['good', 'poor']
pe         | unique:   2 | values: ['no', 'yes']
ane        | unique:   2 | values: ['no', 'yes']
class      | unique:   3 | values: ['ckd', 'ckd\t', 'notckd']
```

```python
# Fixing values in 'dm' and 'class' cols.

for col in df_ckd1.select_dtypes(include ='object').columns:
    df_ckd1[col] = df_ckd1[col].str.strip() # Remove whitespaces, newlines,
         tabs etc.

for col in df_ckd1.select_dtypes(include='object').columns:
    n = df_ckd1[col].nunique()
    vals = sorted(df_ckd1[col].dropna().unique())
    print(f"{col:10} | unique: {n:3} | values: {vals}")
```

```
rbc        | unique:   2 | values: ['abnormal', 'normal']
pc         | unique:   2 | values: ['abnormal', 'normal']
pcc        | unique:   2 | values: ['notpresent', 'present']
ba         | unique:   2 | values: ['notpresent', 'present']
htn        | unique:   2 | values: ['no', 'yes']
dm         | unique:   2 | values: ['no', 'yes']
cad        | unique:   2 | values: ['no', 'yes']
appet      | unique:   2 | values: ['good', 'poor']
pe         | unique:   2 | values: ['no', 'yes']
ane        | unique:   2 | values: ['no', 'yes']
class      | unique:   2 | values: ['ckd', 'notckd']
```

```python
df_ckd1.describe()
```

| | age | bp | sg | al | su | |
|---|---|---|---|---|---|---|
| **count** | 391.000000 | 388.000000 | 353.000000 | 354.000000 | 351.000000 | 356.00 |
| **mean** | 51.483376 | 76.469072 | 1.017408 | 1.016949 | 0.450142 | 148.03 |
| **std** | 17.169714 | 13.683637 | 0.005717 | 1.352679 | 1.099191 | 79.281 |
| **min** | 2.000000 | 50.000000 | 1.005000 | 0.000000 | 0.000000 | 22.000 |
| **25%** | 42.000000 | 70.000000 | 1.010000 | 0.000000 | 0.000000 | 99.000 |

| | age | bp | sg | al | su | |
|---|---|---|---|---|---|---|
| **50%** | 55.000000 | 80.000000 | 1.020000 | 0.000000 | 0.000000 | 121.00 |
| **75%** | 64.500000 | 80.000000 | 1.020000 | 2.000000 | 0.000000 | 163.00 |
| **max** | 90.000000 | 180.000000 | 1.025000 | 5.000000 | 5.000000 | 490.00 |

```
# check each missing value percentage

print(((df_ckd1.isnull().sum()/len(df_ckd1))*100))
```

```
age       2.25
bp        3.00
sg       11.75
al       11.50
su       12.25
rbc      38.00
pc       16.25
pcc       1.00
ba        1.00
bgr      11.00
bu        4.75
sc        4.25
sod      21.75
pot      22.00
hemo     13.00
pcv      17.75
wbcc     26.50
rbcc     32.75
htn       0.50
dm        0.50
cad       0.50
appet     0.25
pe        0.25
ane       0.25
class     0.00
dtype: float64
```

## Preprosseing CKD Dataset

```
# We cant drop Na columns,becuasse most of the field has higher percentage of
         NaN values and will affect the model if we drop it

## Solution : Imputation
from sklearn.impute import SimpleImputer
Numeric_col =
         ['age','bp','sg','al','su','bgr','bu','sod','sc','pot','hemo','pcv','wbcc','rbcc'
Categorical_col=
         ['rbc','pc','pcc','ba','htn','dm','cad','appet','pe','ane','class']

#since from df.describe showed some extream outliers we use Median to impute
num_imputer = SimpleImputer(strategy='median')
df_ckd1[Numeric_col] = num_imputer.fit_transform(df_ckd1[Numeric_col])

#Categorical columns filled with mode
```

```
cat_imputer = SimpleImputer(strategy='most_frequent')
df_ckd1[Categorical_col] =
         cat_imputer.fit_transform(df_ckd1[Categorical_col])

#Check remaining missing values
print(((df_ckd1.isnull().sum()/len(df_ckd1))*100))
```

```
age      0.0
bp       0.0
sg       0.0
al       0.0
su       0.0
rbc      0.0
pc       0.0
pcc      0.0
ba       0.0
bgr      0.0
bu       0.0
sc       0.0
sod      0.0
pot      0.0
hemo     0.0
pcv      0.0
wbcc     0.0
rbcc     0.0
htn      0.0
dm       0.0
cad      0.0
appet    0.0
pe       0.0
ane      0.0
class    0.0
dtype: float64
```

```
# Encoding - We only encode categorical columns

ckd_encoder={}

for col in Categorical_col:
    le = LabelEncoder()
    df_ckd1[col] = le.fit_transform(df_ckd1[col])
    ckd_encoder[col] = le

df_ckd1[Categorical_col]
```

| | rbc | pc | pcc | ba | htn | dm | cad | appet | pe | ane | class |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 |
| **1** | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| **2** | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 0 |
| **3** | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 1 | 1 | 0 |
| **4** | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |

| | rbc | pc | pcc | ba | htn | dm | cad | appet | pe | ane | class |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **395** | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| **396** | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| **397** | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| **398** | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| **399** | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |

400 rows × 11 columns

## Train and Test Data Split - Mushroom Data Set

datasets are split 80/20 using train_test_split() with stratify=y and random_state=42.

5 fold cross-validation is performed using cross_val_score() for all four model-dataset combinations. In 5-fold CV, the training data is divided into 5 equal folds; the model is trained on 4 folds and evaluated on the remaining fold, rotating through all 5 combinations.

```
from sklearn.model_selection import train_test_split

X_m1 = df_m1.drop(columns = ['class'])
Y_m1 = df_m1['class']

X_Mush_train,X_Mush_test,Y_Mush_train,Y_Mush_test = train_test_split(
    X_m1,
    Y_m1,
    test_size=0.2,
    random_state=42,
    stratify=Y_m1
)

print(f"Mushroom Split Result:")
print(f"  X_train: {X_Mush_train.shape}  y_train: {Y_Mush_train.shape}")
print(f"  X_test:  {X_Mush_test.shape}   y_test:  {Y_Mush_test.shape}")
```

```
Mushroom Split Result:
  X_train: (6499, 21)  y_train: (6499,)
  X_test:  (1625, 21)   y_test:  (1625,)
```

## Decision Tree - Mushroom

```
from sklearn.tree import DecisionTreeClassifier, plot_tree
import matplotlib.pyplot as plt

dt_mushroom = DecisionTreeClassifier(
    criterion ='gini',
    max_depth =5,
    min_samples_split = 10,
    min_samples_leaf =4,
    random_state = 42
```

```
)

dt_mushroom.fit(X_Mush_train,Y_Mush_train)
print(f"Mushroom Decision Tree trained.")
print(f"  Actual depth used: {dt_mushroom.get_depth()}")
print(f"  Number of leaves:  {dt_mushroom.get_n_leaves()}")
```

```
Mushroom Decision Tree trained.
  Actual depth used: 5
  Number of leaves:  15
```

```
## Predict

y_predict_dt_mush = dt_mushroom.predict(X_Mush_test)
print("First 20 predictions:", y_predict_dt_mush[:20])
print("First 20 actual:     ", Y_Mush_test.values[:20])
```

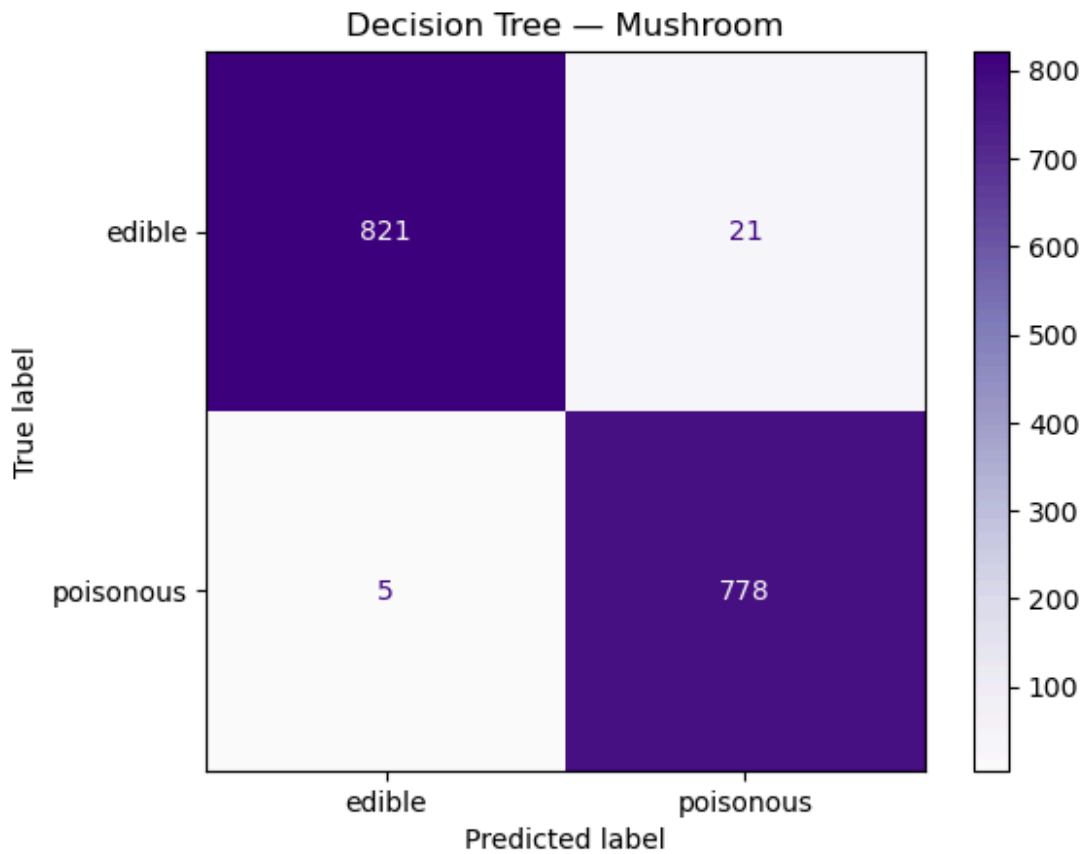
```
First 20 predictions: [1 1 0 1 1 0 0 0 0 1 1 1 0 0 0 1 1 0 0 1]
First 20 actual:      [1 1 0 1 1 0 0 0 0 1 1 1 0 0 0 1 1 0 0 1]
```

```
#Accuracy
from sklearn.metrics import accuracy_score,confusion_matrix,
         ConfusionMatrixDisplay
import matplotlib.pyplot as plt

acc_dt_mush = accuracy_score(Y_Mush_test, y_predict_dt_mush)

print(f"Accuracy: {acc_dt_mush*100:.2f}%")
```

```
Accuracy: 98.40%
```

```
## Confusion Matrix
cm_dt_mush = confusion_matrix(Y_Mush_test, y_predict_dt_mush)
print("\nConfusion Matrix:")
print(cm_dt_mush)

disp = ConfusionMatrixDisplay(cm_dt_mush, display_labels=['edible',
         'poisonous'])
disp.plot(cmap='Purples')
plt.title('Decision Tree – Mushroom')
plt.show()
```

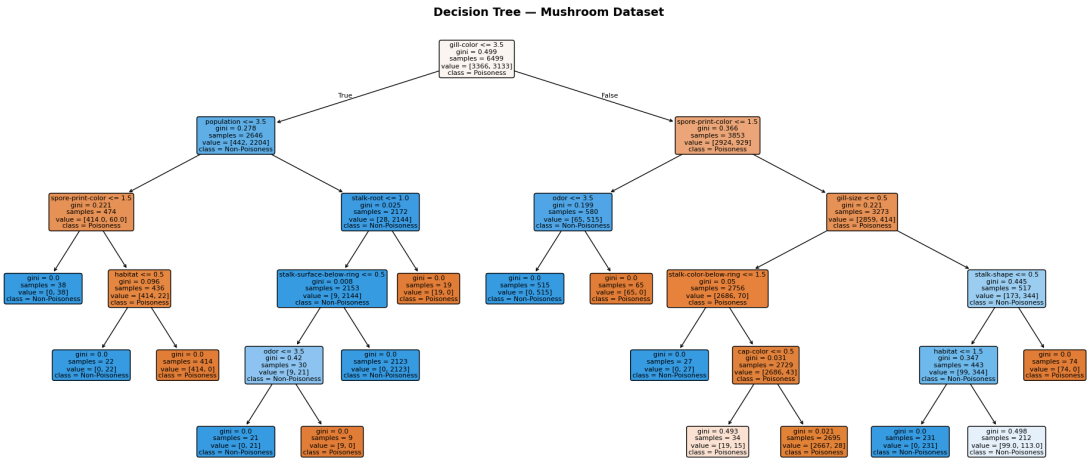
```
Confusion Matrix:
[[821  21]
 [  5 778]]
```




```python
## Classification report

from sklearn.metrics import classification_report
from sklearn.model_selection import cross_val_score

print(classification_report(Y_Mush_test, y_predict_dt_mush,
                            target_names=['edible', 'poisonous']))

# Cross Validation
cv_dt_mush = cross_val_score(dt_mushroom, X_Mush_train, Y_Mush_train, cv=5,
         scoring='accuracy')
print(f"CV Scores: {[round(s*100,1) for s in cv_dt_mush]}")
print(f"CV Mean:   {cv_dt_mush.mean()*100:.2f}%")
print(f"CV Std:    ±{cv_dt_mush.std()*100:.2f}%")
```

```
              precision    recall  f1-score   support

      edible       0.99      0.98      0.98       842
   poisonous       0.97      0.99      0.98       783

    accuracy                           0.98      1625
   macro avg       0.98      0.98      0.98      1625
weighted avg       0.98      0.98      0.98      1625

CV Scores: [np.float64(97.5), np.float64(98.2), np.float64(97.4),
np.float64(97.5), np.float64(98.1)]
CV Mean:   97.71%
CV Std:    ±0.33%
```

```
## Visualising the DT

plt.figure(figsize=(24, 10))
plot_tree(
    dt_mushroom,
    feature_names=X_m1.columns.tolist(),
    class_names=['Poisoness', 'Non-Poisoness'],
    filled=True,
    rounded=True,
    fontsize=8
)
plt.title('Decision Tree — Mushroom Dataset', fontsize=14, fontweight='bold')
plt.savefig('dt_Mushroom.png', dpi=150, bbox_inches='tight')
plt.show()
```




## Train and Test Data Split - CKD Data Set

```
X_Ckd = df_ckd1.drop(columns='class')
Y_Ckd = df_ckd1['class']

X_ckd_Train,X_ckd_Test,Y_ckd_Train,Y_ckd_Test = train_test_split(
    X_Ckd,
    Y_Ckd,
    test_size = 0.2,
    random_state = 42,
    stratify = Y_Ckd
)

print(f"CKD Split Result:")
print(f"  X_train: {X_ckd_Train.shape}  y_train: {Y_ckd_Train.shape}")
print(f"  X_test:  {X_ckd_Test.shape}   y_test:  {Y_ckd_Test.shape}")
```

```
CKD Split Result:
  X_train: (320, 24)  y_train: (320,)
  X_test:  (80, 24)   y_test:  (80,)
```

## Decision Tree - CKD

```
dt_ckd = DecisionTreeClassifier(
    criterion ='gini',
    max_depth =5,
    min_samples_split = 10,
    min_samples_leaf =4,
    random_state = 42

)

dt_ckd.fit(X_ckd_Train,Y_ckd_Train)
print(f"CKD Decision Tree trained.")
print(f"  Actual depth used: {dt_ckd.get_depth()}")
print(f"  Number of leaves:  {dt_ckd.get_n_leaves()}")
```

```
CKD Decision Tree trained.
  Actual depth used: 5
  Number of leaves:  8
```

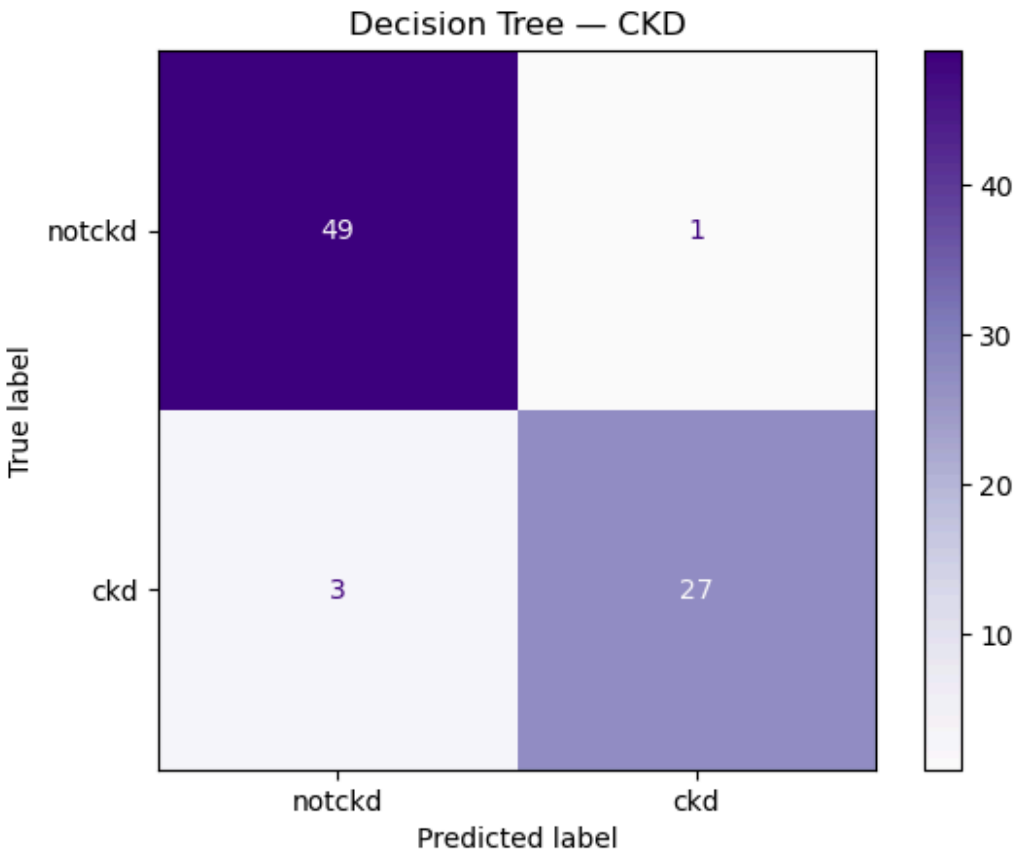
```
## Predict

y_predict_dt_ckd = dt_ckd.predict(X_ckd_Test)
print("First 20 predictions:", y_predict_dt_ckd[:20])
print("First 20 actual:     ", Y_ckd_Test.values[:20])
```

```
First 20 predictions: [0 0 1 1 1 0 0 0 1 0 0 0 1 0 1 0 0 1 0 1]
First 20 actual:      [0 0 1 1 1 0 0 0 1 0 0 0 1 0 1 0 0 1 0 1]
```

```
## Accuracy, COnfusion Matrix and Classification Report

acc_dt_ckd = accuracy_score(Y_ckd_Test, y_predict_dt_ckd)

print(f"Accuracy: {acc_dt_ckd*100:.2f}%")

#Confusion Matrix
cm_dt_ckd = confusion_matrix(Y_ckd_Test, y_predict_dt_ckd)
print("\nConfusion Matrix:")
print(cm_dt_ckd)

disp = ConfusionMatrixDisplay(cm_dt_ckd, display_labels=['notckd', 'ckd'])
disp.plot(cmap='Purples')
plt.title('Decision Tree – CKD')
plt.show()

## Classification report
print(classification_report(Y_ckd_Test, y_predict_dt_ckd,
                            target_names=['notckd', 'ckd']))

# Cross Validation
cv_dt_ckd = cross_val_score(dt_ckd, X_ckd_Train, Y_ckd_Train, cv=5,
         scoring='accuracy')
print(f"CV Scores: {[round(s*100,1) for s in cv_dt_ckd]}")
```

```
print(f"CV Mean:   {cv_dt_ckd.mean()*100:.2f}%")
print(f"CV Std:    ±{cv_dt_ckd.std()*100:.2f}%")
```

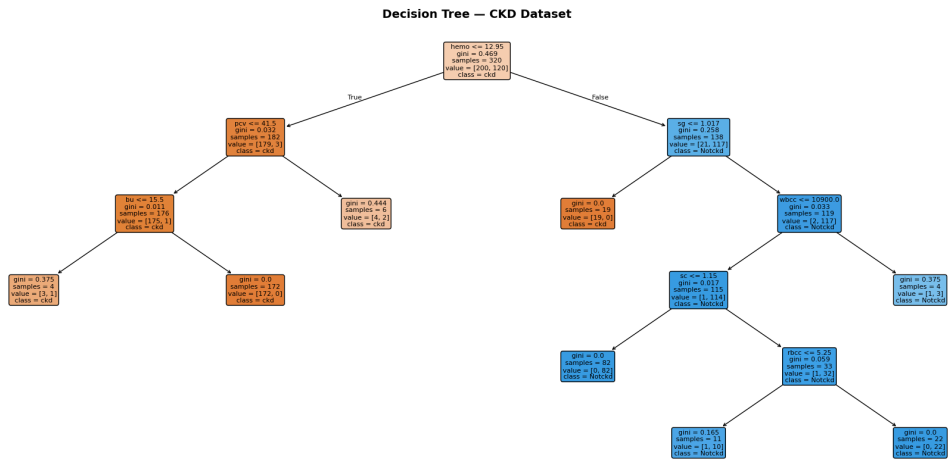
```
Accuracy: 95.00%

Confusion Matrix:
[[49  1]
 [ 3 27]]
```




```
              precision    recall  f1-score   support

      notckd       0.94      0.98      0.96        50
         ckd       0.96      0.90      0.93        30

    accuracy                           0.95        80
   macro avg       0.95      0.94      0.95        80
weighted avg       0.95      0.95      0.95        80

CV Scores: [np.float64(96.9), np.float64(92.2), np.float64(96.9),
np.float64(100.0), np.float64(98.4)]
CV Mean:   96.88%
CV Std:    ±2.61%
```

```
## Visualising the DT

plt.figure(figsize=(24, 10))
plot_tree(
    dt_ckd,
    feature_names=X_Ckd.columns.tolist(),
```

```
    class_names=['ckd', 'Notckd'],
    filled=True,
    rounded=True,
    fontsize=8
)
plt.title('Decision Tree — CKD Dataset', fontsize=14, fontweight='bold')
plt.savefig('dt_ckd.png', dpi=150, bbox_inches='tight')
plt.show()
```

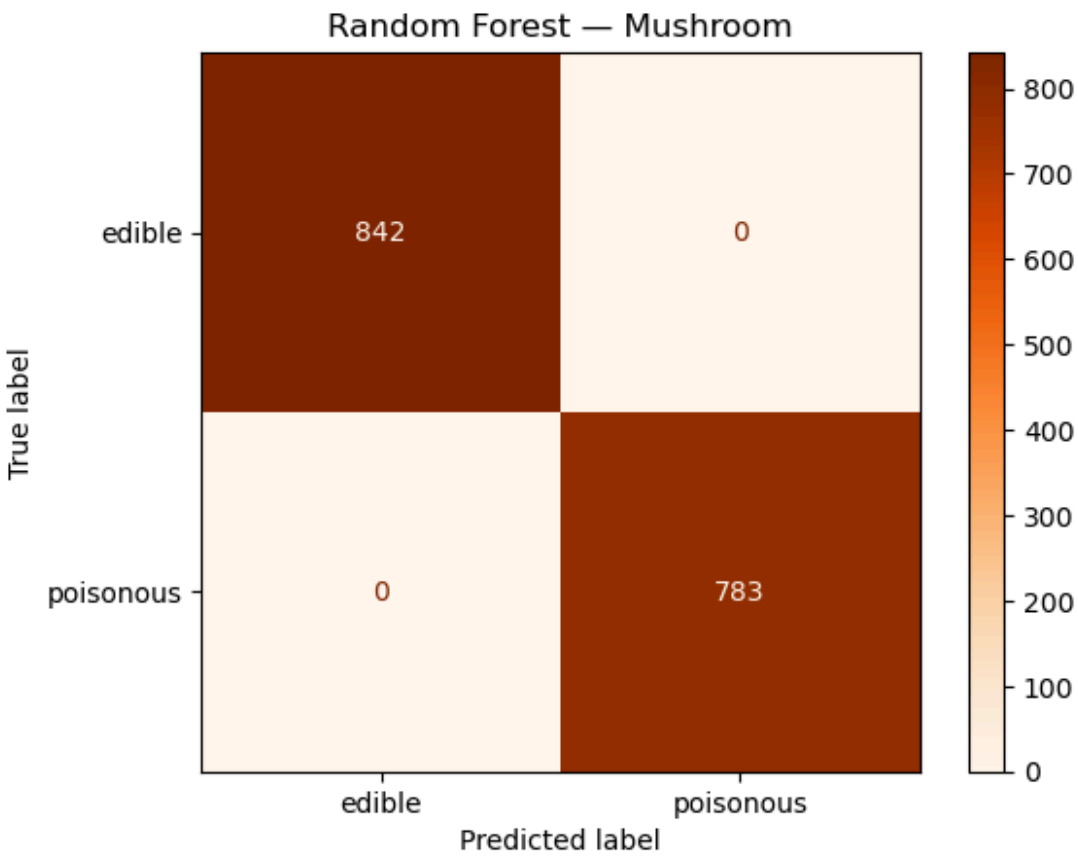



## Random Forest on Mushroom Data set

```
from sklearn.ensemble import RandomForestClassifier

rf_mush = RandomForestClassifier(
    n_estimators=100,
    max_depth=15,
    min_samples_split=5,
    min_samples_leaf=2,
    max_features='sqrt',
    random_state=42,
    n_jobs=-1
)

rf_mush.fit(X_Mush_train, Y_Mush_train)
y_pred_rf_mush = rf_mush.predict(X_Mush_test)

print("First 10 predictions:", y_pred_rf_mush[:10])
print("First 10 actual:     ", Y_Mush_test.values[:10])

#Accuracy
acc_rf_mush = accuracy_score(Y_Mush_test, y_pred_rf_mush)
print(f"\nAccuracy: {acc_rf_mush*100:.2f}%")

#Confusion Matrix
cm_rf_mush = confusion_matrix(Y_Mush_test, y_pred_rf_mush)
print("\nConfusion Matrix:")
print(cm_rf_mush)
```

```python
disp = ConfusionMatrixDisplay(cm_rf_mush, display_labels=['edible',
         'poisonous'])
disp.plot(cmap='Oranges')
plt.title('Random Forest — Mushroom')
plt.show()

#Classification Report
print(classification_report(Y_Mush_test, y_pred_rf_mush,
                            target_names=['edible', 'poisonous']))

#Cross Validation
cv_rf_mush = cross_val_score(rf_mush, X_Mush_train, Y_Mush_train, cv=5,
         scoring='accuracy')
print(f"CV Scores: {[round(s*100,1) for s in cv_rf_mush]}")
print(f"CV Mean:   {cv_rf_mush.mean()*100:.2f}%")
print(f"CV Std:    ±{cv_rf_mush.std()*100:.2f}%")
```

```
First 10 predictions: [1 1 0 1 1 0 0 0 0 1]
First 10 actual:      [1 1 0 1 1 0 0 0 0 1]

Accuracy: 100.00%

Confusion Matrix:
[[842   0]
 [  0 783]]
```




```
              precision    recall  f1-score   support

      edible       1.00      1.00      1.00       842
```

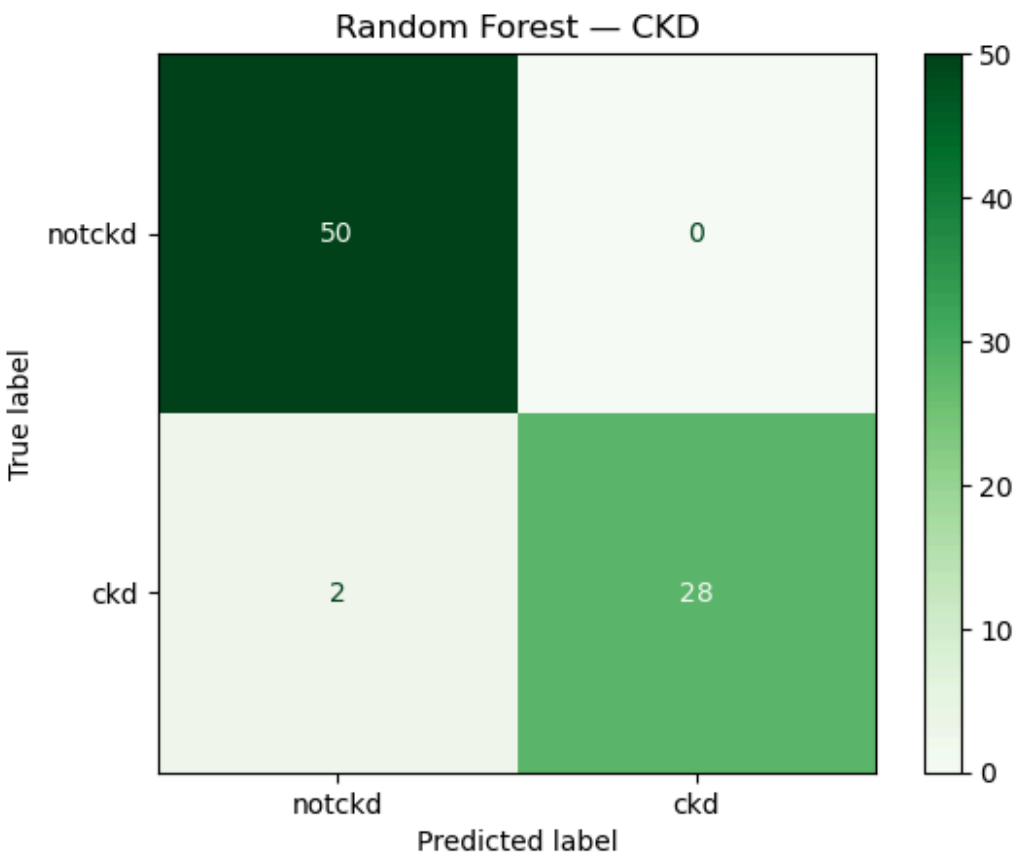
```
   poisonous       1.00      1.00      1.00       783

    accuracy                           1.00      1625
   macro avg       1.00      1.00      1.00      1625
weighted avg       1.00      1.00      1.00      1625

CV Scores: [np.float64(99.7), np.float64(100.0), np.float64(100.0),
np.float64(100.0), np.float64(100.0)]
CV Mean:   99.94%
CV Std:    ±0.12%
```

## Random Forest on CKD Data Set

```python
# Train
rf_ckd = RandomForestClassifier(
    n_estimators=100,
    max_depth=15,
    min_samples_split=5,
    min_samples_leaf=2,
    max_features='sqrt',
    random_state=42,
    n_jobs=-1
)

rf_ckd.fit(X_ckd_Train, Y_ckd_Train)
# Predict
y_pred_rf_ckd = rf_ckd.predict(X_ckd_Test)

print("First 10 predictions:", y_pred_rf_ckd[:10])
print("First 10 actual:     ", Y_ckd_Test.values[:10])

# Accuracy
acc_rf_ckd = accuracy_score(Y_ckd_Test, y_pred_rf_ckd)
print(f"\nAccuracy: {acc_rf_ckd*100:.2f}%")

# Confusion Matrix
cm_rf_ckd = confusion_matrix(Y_ckd_Test, y_pred_rf_ckd)
print("\nConfusion Matrix:")
print(cm_rf_ckd)

disp = ConfusionMatrixDisplay(cm_rf_ckd, display_labels=['notckd', 'ckd'])
disp.plot(cmap='Greens')
plt.title('Random Forest – CKD')
plt.show()

#  Classification Report
print(classification_report(Y_ckd_Test, y_pred_rf_ckd,
                            target_names=['notckd', 'ckd']))

# Cross Validation
cv_rf_ckd = cross_val_score(rf_ckd, X_ckd_Train, Y_ckd_Train, cv=5,
         scoring='accuracy')
print(f"CV Scores: {[round(s*100,1) for s in cv_rf_ckd]}")
```

```
print(f"CV Mean:   {cv_rf_ckd.mean()*100:.2f}%")
print(f"CV Std:    ±{cv_rf_ckd.std()*100:.2f}%")
```

```
First 10 predictions: [0 0 1 1 1 0 0 0 1 0]
First 10 actual:      [0 0 1 1 1 0 0 0 1 0]

Accuracy: 97.50%

Confusion Matrix:
[[50  0]
 [ 2 28]]
```




```
              precision    recall  f1-score   support

      notckd       0.96      1.00      0.98        50
         ckd       1.00      0.93      0.97        30

    accuracy                           0.97        80
   macro avg       0.98      0.97      0.97        80
weighted avg       0.98      0.97      0.97        80

CV Scores: [np.float64(98.4), np.float64(98.4), np.float64(98.4),
np.float64(100.0), np.float64(98.4)]
CV Mean:   98.75%
CV Std:    ±0.62%
```

from sklearn.metrics import f1_score

def show_summary(models_info): print(" OVERALL MODEL COMPARISON SUMMARY") print("="*50) print(f" {'Model':<16} {'Dataset':<10} {'Test Acc':>10} " f" {'F1':>8}{'False Positive':>20}{'False Negative':>20}{'CV Mean':>10} {'CV Std':>8}") print(" " + "-"*150)

```
for m in models_info:
    print(f"  {m['name']:<16} {m['dataset']:<10} "
          f"{m['acc']*100:>9.2f}% "
          f"{m['f1']:>8.4f} "
          f"{m['cm_fp']:>20} "
          f"{m['cm_fn']:>20} "
          f"{m['cv_mean']*100:>9.2f}% "
          f"±{m['cv_std']*100:>5.2f}%")

# Find best model
best = max(models_info, key=lambda x: x['acc'])
print(f"\n Best Model: {best['name']}")
print(f"   Accuracy: {best['acc']*100:.2f}%  |  F1: {best['f1']:.4f}")
```

models_info = [ { 'name': 'Decision Tree', 'dataset': 'CKD', 'acc': acc_dt_ckd, 'f1': f1_score(Y_ckd_Test, y_predict_dt_ckd, average='weighted'), 'cv_mean': cv_dt_ckd.mean(), 'cv_std': cv_dt_ckd.std(), 'model': dt_ckd, 'cm_fp': cm_dt_ckd[0][1], 'cm_fn': cm_dt_ckd[1][0]

```
},
{
    'name':    'Random Forest',
    'dataset': 'CKD',
    'acc':     acc_rf_ckd,
    'f1':      f1_score(Y_ckd_Test,  y_pred_rf_ckd,  average='weighted'),
    'cv_mean': cv_rf_ckd.mean(),
    'cv_std':  cv_rf_ckd.std(),
    'model':   rf_ckd,
    'cm_fp':   cm_rf_ckd[0][1],
    'cm_fn':   cm_rf_ckd[1][0]
},
{
    'name':    'Decision Tree',
    'dataset': 'Mushroom',
    'acc':     acc_dt_mush,
    'f1':      f1_score(Y_Mush_test, y_predict_dt_mush, average='weighted'),
    'cv_mean': cv_dt_mush.mean(),
    'cv_std':  cv_dt_mush.std(),
    'model':   dt_mushroom,
    'cm_fp':   cm_dt_mush[0][1],
    'cm_fn':   cm_dt_mush[1][0]
},
{
    'name':    'Random Forest',
    'dataset': 'Mushroom',
    'acc':     acc_rf_mush,
    'f1':      f1_score(Y_Mush_test, y_pred_rf_mush, average='weighted'),
    'cv_mean': cv_rf_mush.mean(),
    'cv_std':  cv_rf_mush.std(),
    'model':   rf_mush,
    'cm_fp':   cm_rf_mush[0][1],
```

```
    'cm_fn':   cm_rf_mush[1][0]
},

]
```

show_summary(models_info)

## The Random Forest shows higher metrics than the Decision Tree on all four evaluation metrics across both datasets.

**On CKD, the Random Forest achieved 97.50% accuracy and F1=0.9748, while the Decision Tree achieved 95.00% and F1=0.9496, with tighter cross-validation stability (±0.62% vs ±2.61%). Further, the Random Forest reduced False Negatives on the CKD test set from 3 to 2, provided that one fewer patient with kidney disease was missed. This is very critical in sensitive data prediction, like medical diagnosis. On Mushroom, the Random Forest achieved perfect classification (100% accuracy, F1=1.0, zero False Negatives) versus 5 False Negatives for the Decision Tree. No evidence of overfitting was observed in either model, as cross-validation scores closely matched test set performance across all folds.**

## Exporting best model

```python
import pickle

# Initialising the best models for the 2 datasets

best_ckd = rf_ckd
best_mush = rf_mush

ckd_file = 'best_ckd_model.pkl'
mush_file  = 'best_mush_model.pkl'

file = open(ckd_file,'wb')
pickle.dump(best_ckd,file)
file.close()
print("CKD Model exported")

file = open(mush_file,'wb')
pickle.dump(best_mush,file)
file.close()
print("Mush Model exported")
```

```
CKD Model exported
Mush Model exported
```